

Here is the **complete Packer template** that provisions a VM with JupyterHub, `configurable-http-proxy`, pulls SSL certificates from Google Cloud Storage (GCS), and fetches software from Nexus using credentials provided at runtime. This version uses **environment variables** for Nexus credentials, and sets up **systemd services** for JupyterHub and the proxy.

Packer Template:

```
``json
{
  "variables": {
    "project_id": "<your-gcp-project-id>",
    "zone": "us-central1-a",
    "image_family": "debian-11",
    "jupyterhub_version": "1.0.0",
    "nexus_username": "{{env `NEXUS_USERNAME`}}",
    "nexus_password": "{{env `NEXUS_PASSWORD`}}",
    "gcs_bucket": "your-bucket-name"
  },

  "builders": [
    {
      "type": "googlecompute",
      "project_id": "{{user `project_id`}}",
      "zone": "{{user `zone`}}",
      "source_image_family": "{{user `image_family`}}",
      "instance_name": "jupyterhub-{{timestamp}}",
      "disk_size": 10,
      "image_name": "jupyterhub-{{timestamp}}",
      "image_family": "jupyterhub-images",
      "metadata": {
        "enable-oslogin": "TRUE"
      }
    }
  ],

  "provisioners": [
    {
      "type": "shell",
      "inline": [
        "sudo apt-get update",
        "sudo apt-get install -y python3-pip npm nodejs docker.io",
        "sudo systemctl start docker",
        "sudo systemctl enable docker",
```

```

        "sudo pip3 install jupyterhub notebook",
        "sudo npm install -g configurable-http-proxy",
        "sudo pip3 install oauthenticator",
        "sudo pip3 install samlauthenticator"
    ]
},
{
    "type": "shell",
    "inline": [
        "sudo apt-get install -y google-cloud-sdk",
        "gcloud auth activate-service-account --key-file ~/jupyterhub-sa-key.json",
        "gsutil cp gs://{user `gcs_bucket`} /fullchain.pem /etc/ssl/certs/fullchain.pem",
        "gsutil cp gs://{user `gcs_bucket`} /privkey.pem /etc/ssl/private/privkey.pem"
    ]
},
{
    "type": "shell",
    "inline": [
        "wget --user={{user `nexus_username`} --password={{user `nexus_password`}}
<NEXUS_REPO_URL>/path/to/your/software",
        "sudo dpkg -i software.deb"
    ]
},
{
    "type": "file",
    "source": "jupyterhub_config.py",
    "destination": "/etc/jupyterhub/jupyterhub_config.py"
},
{
    "type": "file",
    "source": "azure_ad_metadata.xml",
    "destination": "/etc/jupyterhub/azure_ad_metadata.xml"
},
{
    "type": "shell",
    "inline": [
        "cat <<EOF | sudo tee /etc/systemd/system/jupyterhub.service",
        "[Unit]",
        "Description=JupyterHub",
        "After=network.target",
        "",
        "[Service]",
        "User=<your-username>",
        "ExecStart=/usr/local/bin/jupyterhub -f /etc/jupyterhub/jupyterhub_config.py",

```

```

    "Restart=always",
    "WorkingDirectory=/etc/jupyterhub",
    "",
    "[Install]",
    "WantedBy=multi-user.target",
    "EOF"
  ]
},
{
  "type": "shell",
  "inline": [
    "cat <<EOF | sudo tee /etc/systemd/system/configurable-http-proxy.service",
    "[Unit]",
    "Description=Configurable HTTP Proxy",
    "After=network.target",
    "",
    "[Service]",
    "ExecStart=/usr/local/bin/configurable-http-proxy --ip 0.0.0.0 --port 8000 --ssl-key",
    "/etc/ssl/private/privkey.pem --ssl-cert /etc/ssl/certs/fullchain.pem",
    "Restart=always",
    "",
    "[Install]",
    "WantedBy=multi-user.target",
    "EOF"
  ]
},
{
  "type": "shell",
  "inline": [
    "sudo systemctl daemon-reload",
    "sudo systemctl enable jupyterhub",
    "sudo systemctl start jupyterhub",
    "sudo systemctl enable configurable-http-proxy",
    "sudo systemctl start configurable-http-proxy",
    "echo 'Setting up GKE connection...'",
    "gcloud auth activate-service-account --key-file ~/jupyterhub-sa-key.json",
    "gcloud container clusters get-credentials <your-gke-cluster-name> --zone",
    "<your-gke-zone>"
  ]
}
]
}
...

```

Breakdown of the Template:

1. **Variables**:

- You'll need to pass your GCP project, Nexus credentials, GCS bucket, and other necessary details.
- **NEXUS_USERNAME** and **NEXUS_PASSWORD** are expected to be passed as environment variables before running the build.

2. **Builders**:

- The VM will be created in GCP, using Debian 11 as the base image.

3. **Provisioners**:

- **Install Required Packages**: Installs JupyterHub, `configurable-http-proxy`, and required packages.
- **Download SSL Certificates**: Fetches SSL certificates from GCS.
- **Download Software from Nexus**: Fetches software from Nexus using the credentials passed via environment variables.
- **Copy Configuration Files**: Copies JupyterHub configuration and Azure AD SAML metadata.
- **Create systemd Services**: Creates and enables services for JupyterHub and the proxy.
- **GKE Cluster Setup**: Authenticates with GKE using a service account and sets up credentials for interacting with Kubernetes.

How to Use It:

1. Export Nexus Credentials:

```
```bash
export NEXUS_USERNAME="your-nexus-username"
export NEXUS_PASSWORD="your-nexus-password"
```
```

2. Build the Image with Packer:

```
```bash
packer build your-packer-template.json
```
```

3. After Packer Finishes:

The generated image will contain JupyterHub and all required services ready to be started on boot. The `systemd` services ensure that both **JupyterHub** and **configurable-http-proxy** will automatically start when the VM is created.

Important Details:

- **GCS SSL Certs**: The certificates are pulled from your GCS bucket. Ensure that your GCS bucket is properly configured, and the service account (`jupyterhub-sa-key.json`) has the necessary access.

- **Nexus Credentials**: Nexus username and password are passed via environment variables to prevent them from being hard-coded in the template.

- **Systemd Services**: The template configures two system services (`jupyterhub` and `configurable-http-proxy`), which handle the lifecycle of JupyterHub and the proxy.

Let me know if you have any specific questions about the code!